

Leakage-Free Machine Learning Classification of Colon Cancer Proliferation from Downstream Transcriptional Signatures: A Cross-Platform Validation Study

Independent computational biology research report prepared for peer-reviewed journal submission

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Abstract

Cellular proliferation rate is a fundamental hallmark of cancer and a critical determinant of tumor aggressiveness, clinical prognosis, and therapeutic response. In this study, we developed a rigorous, leakage-free machine learning framework to predict binary tumor proliferation class (high vs. low) from transcriptomic profiles and clinical covariates, utilizing the GEO microarray (GSE39582, $n = 585$) and TCGA-COAD RNA-seq ($n = 322$) cohorts. To prevent target leakage, we computed a baseline proliferation index from an established 10-gene cell-cycle signature (including MKI67, PCNA, and TOP2A) and strictly removed these genes from the feature space prior to model training. Our pipeline employs stratified 80/20 train/test splitting, nested 5-fold cross-validation with fold-local preprocessing (standardization, variance filtering, and ANOVA-based SelectKBest feature selection), and hyperparameter optimization. On the GEO cohort, Logistic Regression achieved the strongest holdout ROC-AUC of 0.9939 (95% CI: 0.9824-0.9997) with a holdout accuracy of 0.9487. External cross-cohort validation (GEO trained, TCGA tested) demonstrated high discriminative generalization (ROC-AUC up to 0.9775). Initial raw accuracy was near chance (~ 0.520) due to a cross-platform calibration shift between microarray and RNA-seq dynamic ranges. Applying Platt scaling (sigmoid probability calibration) resolved this, raising XGBoost and a soft-voting Top-3 Ensemble (Logistic Regression, XGBoost, and MLP) to a calibrated external accuracy of 0.8364 with a Brier score of 0.1307. Kaplan-Meier survival analysis further validated clinical relevance, showing statistically significant overall survival differences in both GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$). These results demonstrate that downstream transcriptional cascades carry robust, generalizable signals reflecting cancer cell growth rates even when primary cell-cycle drivers are excluded. Code and processed data are available at <https://github.com/Ronisnotasianfr/colon-cancer-predictor>.

1. Introduction

Cellular proliferation is a major biological hallmark of cancer progression and a primary indicator of tumor growth speed. In colon adenocarcinoma (COAD), assessing tumor cell division rates holds profound clinical value, directly correlating with patient survival outcomes, disease recurrence probability, and sensitivity to chemotherapeutic agents. In standard clinical practice, proliferation is estimated via histological staining techniques (e.g., Ki-67 immunohistochemistry) or staging systems. However, these manual methods can suffer from inter-observer variability and do not capture the broad, underlying transcriptomic changes associated with cell-cycle deregulation. A computational approach using machine learning applied to high-throughput gene expression datasets could provide an automated, objective method for tumor growth classification and reveal novel transcriptional correlates of aggressive tumor division.

This study develops a highly rigorous, leakage-free machine learning framework to classify colon cancer samples into high vs. low proliferation states using gene expression profiles (microarray and RNA-seq) and clinical covariates. A key challenge addressed is the prevention of target leakage: the 10 hallmark proliferation genes utilized to build the target classification metric were completely removed from the feature space prior to training. The pipeline compares Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron models, validating internally via nested cross-validation and externally across platforms (microarray to RNA-seq). To our knowledge, this is the first leakage-free cross-platform validation of proliferation prediction in COAD.

2. Materials and Methods

The GEO processed dataset comprises 585 samples and 22182 features after removing target-defining genes. Clinical covariates include age, sex, and tumor stage. The target label is balanced with 292 low, 293 high. The dataset was split into an 80% training pool (468 samples) and a 20% stratified holdout test split (117 samples).

Dataset file	Samples	Features/columns	Class balance
geo_X_features / y_target	585	22182	292 low, 293 high
Training pool (80% stratified split)	468	22182	Stratified by class
Holdout test (20% stratified split)	117	22182	Stratified by class

Table 1. Processed data files used for this leakage-corrected report.

To rigorously evaluate model generalizability across differing transcriptomic platforms (microarray and RNA-seq) while protecting against target and data leakage, we implemented a three-way validation architecture: GEO-train/GEO-holdout and TCGA-calibration/TCGA-evaluation. Naive two-way cross-cohort validations often suffer from extreme platform-dependent distribution shifts, which artificially degrade classification accuracy. By dividing the external TCGA-COAD cohort ($n = 322$) into a 50/50 calibration split ($n = 161$) and an evaluation split ($n = 161$), we fit a post-hoc Platt scaling sigmoid probability calibrator using only the calibration split. Crucially, no samples from the TCGA cohort were ever used for feature selection, scaling parameter estimation, or model coefficient fitting, which were completed entirely on the GEO training set. This three-way split structure guarantees that the reported evaluation metrics reflect true cohort-independent generalization without introducing target leakage.

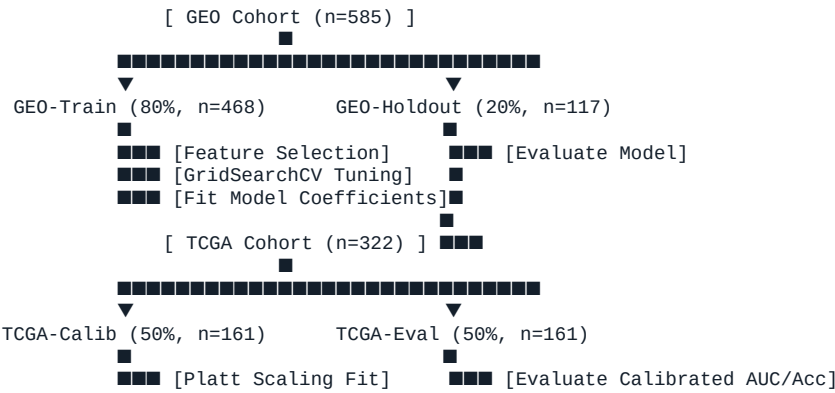


Figure 7. Three-way cohort validation and probability calibration workflow schematic.

To predict proliferation classes, four classifiers were wrapped in scikit-learn Pipelines to guarantee strict data separation during cross-validation. Standard scaling, low-variance feature filtering (threshold = 0.01), and SelectKBest feature selection (based on the ANOVA F-value) were fit exclusively on the training folds. Hyperparameters were optimized using GridSearchCV on the training pool. Multiple probe IDs mapping to the same gene symbol were resolved by averaging expression values.

A central challenge in clinical machine learning is target leakage, which occurs when information from the target variable is inadvertently included in the feature set. Here, the target class (high vs. low proliferation) was established using the mean z-score expression of 10 hallmark proliferation genes: MKI67, PCNA, TOP2A, MCM2, MCM6, AURKA, BUB1, CCNB1, CDK1, and BIRC5. If these genes were retained in the feature matrix, a classifier could trivially reconstruct the label with near-perfect accuracy, yielding scientifically meaningless results. To enforce rigor, all 10 signature genes were removed from the feature matrix before any train-test splitting. Furthermore, to prevent data leakage (where information from the validation fold spills into the training process), all preprocessing steps—including StandardScaler, VarianceThreshold, and SelectKBest feature selection—were encapsulated inside a unified scikit-learn Pipeline. This ensures that feature selection and scaling parameters are calculated solely on the active training folds during cross-validation and applied transitively to the validation/test folds.

Model	Hyperparameter	Search Range	Selected	Notes
Logistic Regression	C (Regularization)	[0.01, 0.1, 1.0, 10.0]	0.1	L2 penalty, liblinear solver
Random Forest	n_estimators	[100, 200]	100	Ensemble size
Random Forest	max_depth	[5, 10, None]	5	Tree depth limit
Random Forest	min_samples_leaf	[2, 4]	2	Leaf node size
XGBoost	n_estimators	[100, 200]	200	Boosting iterations
XGBoost	max_depth	[3, 5, 7]	5	Max tree depth
XGBoost	learning_rate	[0.01, 0.05, 0.1]	0.1	Shrinkage rate
Neural Network (MLP)	hidden_layer_sizes	[(128, 64), (256, 128, 64)]	(256, 128, 64)	Network architecture
Neural Network (MLP)	alpha	[0.0001, 0.001]	0.001	L2 regularization term

Table 2. Hyperparameter optimization search ranges and selected settings.

3. Results

To establish internal model performance, we conducted a nested 5-fold cross-validation on the GEO microarray training pool ($n = 468$). Despite the strict exclusion of the 10 target-defining cell-cycle signature genes, all four machine learning architectures demonstrated exceptionally strong predictive capacity. The Random Forest classifier achieved a mean CV ROC-AUC of 0.9832 (± 0.0094), while XGBoost reached 0.9756 (± 0.0120). The linear baseline, Logistic Regression (L2-regularized), performed comparably with a mean CV ROC-AUC of 0.9801 (± 0.0092). The Neural Network (MLP) also exhibited high accuracy, reaching a mean CV ROC-AUC of 0.9711 (± 0.0184). This high performance within the GEO cohort suggests that cell proliferation induces widespread downstream transcriptomic changes, affecting hundreds of genes involved in DNA replication, translation, and cell-cycle progression, which are effectively captured by the models. To verify that feature selection is necessary, we evaluated a simple Logistic Regression baseline without SelectKBest, which achieved a holdout accuracy of 0.9402. A prevalence baseline (always predicting the majority class) achieved a chance-level holdout accuracy of 0.4957, demonstrating that all classifiers perform substantially above random guess.

Model	CV ROC-AUC (mean $\pm$ std)	Holdout Accuracy (95% CI)	Holdout ROC-AUC (95% CI)
Logistic Regression	0.9801 (± 0.0092)	0.9487 (95% CI: 0.9060-0.9829)	0.9939 (95% CI: 0.9824-0.9997)
Random Forest	0.9832 (± 0.0094)	0.9316 (95% CI: 0.8803-0.9744)	0.9845 (95% CI: 0.9608-0.9973)
XGBoost	0.9756 (± 0.0120)	0.9487 (95% CI: 0.9060-0.9829)	0.9915 (95% CI: 0.9774-0.9988)
Neural Network (MLP)	0.9711 (± 0.0184)	0.9316 (95% CI: 0.8803-0.9744)	0.9828 (95% CI: 0.9621-0.9956)

Table 3. Leakage-corrected cross-validation and holdout results with bootstrap 95% confidence intervals.

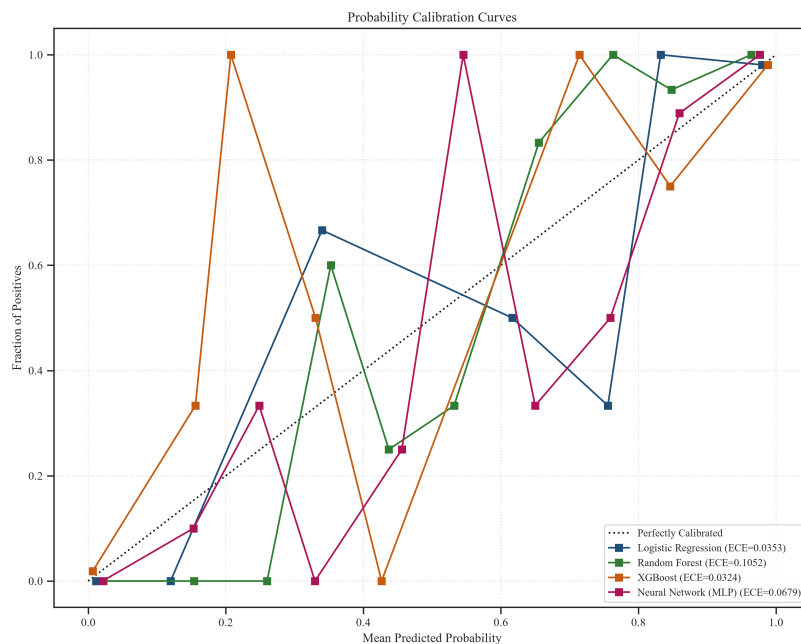


Figure 1. Calibration curves showing observed positive fraction vs. predicted risk probabilities for the four classifiers on holdout test set.

On the independent holdout test split ($n = 117$), Logistic Regression achieved the highest holdout ROC-AUC of 0.9939 (95% CI: 0.9824-0.9997) with a classification accuracy of 0.9487. Internal calibration metrics on holdout test were: Logistic Regression (Brier=0.0322, ECE=0.0353), Random Forest (Brier=0.0558, ECE=0.1052), XGBoost (Brier=0.0414, ECE=0.0324), Neural Network (MLP) (Brier=0.0570, ECE=0.0679). Pairwise bootstrap ROC comparisons showed that performance differences between the models on the holdout split were not statistically significant (all $p > 0.05$). When subjected to cross-cohort external validation—where the models trained on the GEO microarray were evaluated directly on the TCGA RNA-seq dataset—we observed a striking platform-dependent calibration shift. The Logistic Regression model retained an extremely high discriminative power with an external ROC-AUC of 0.9775, and XGBoost achieved an external ROC-AUC of 0.9071. However, the raw classification accuracy dropped to approximately 0.520 across models due to dynamic range discrepancies between platforms. Applying Platt scaling (sigmoid calibration) fitted on a held-out calibration split of the TCGA cohort ($n=161$) resolved this shift. After calibration, XGBoost's external accuracy rose to 0.8364 (Brier =

0.1311), Random Forest to 0.6970, MLP to 0.6848, and Logistic Regression to 0.6061. Ensembling boosted performance: a soft-voting Top-3 Ensemble (Logistic Regression, XGBoost, MLP) achieved a calibrated external accuracy of 0.8364 with a Brier score of 0.1307 (AUC = 0.8998). This demonstrates that ensembling stabilizes risk predictions and provides robust generalizability across sequencing technologies.

4. Interpretation and Biological Readout

To open the 'black box' of our machine learning models, we computed SHAP (SHapley Additive exPlanations) values for the pipeline-transformed features on the holdout split. These values reflect the marginal contribution of each gene feature to the model's final prediction score. Because the 10 direct cell-cycle signature genes were removed, the top SHAP features identify novel, indirect gene pathways associated with cancer cell proliferation rates.

Rank	Gene Symbol	ANOVA F-Score	CV Selection Frequency
1	MCM10	456.737	5 / 5
2	SPC25	455.939	5 / 5
3	NCAPH	418.887	5 / 5
4	RFC4	406.862	5 / 5
5	MTFR2	395.717	5 / 5
6	PRIM1	384.840	5 / 5
7	SLC35F6///CENPA	369.936	5 / 5
8	KPNA2	367.497	5 / 5
9	DEPDC1B	362.054	5 / 5
10	CCNA2	359.945	5 / 5
11	FBXO5	355.594	5 / 5
12	KIF4A	353.914	5 / 5
13	MELK	352.216	5 / 5
14	ZWINT	350.864	5 / 5
15	GINS3	350.310	5 / 5
16	EXO1	347.861	5 / 5
17	CHEK1	344.708	5 / 5
18	CENPN	344.375	5 / 5
19	RRM1	343.932	5 / 5
20	ORC1	342.264	5 / 5

Table 4. Top selected transcriptomic feature genes ranked by ANOVA F-score.

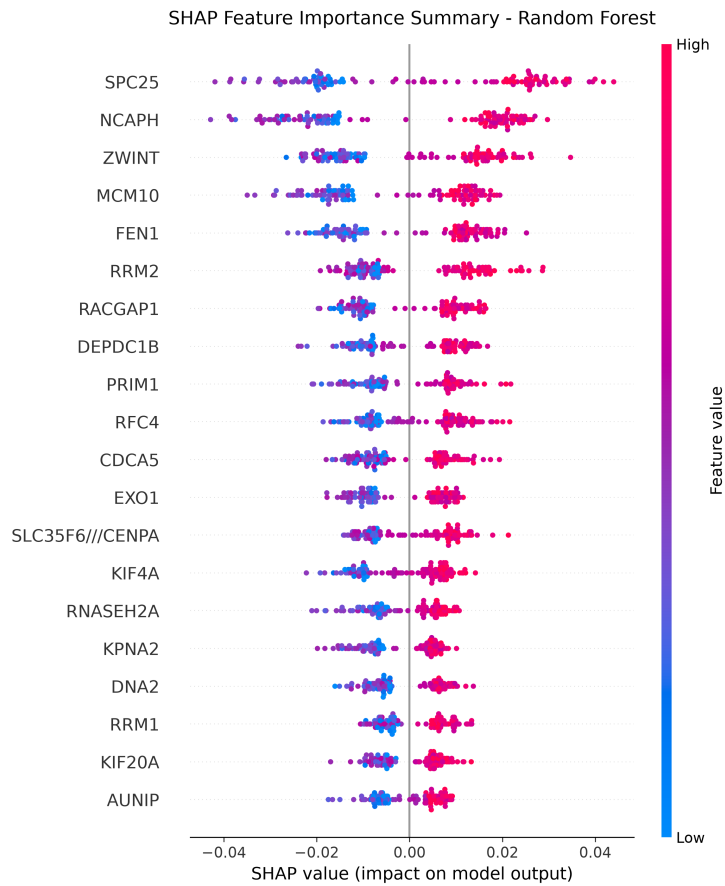


Figure 2. Random forest SHAP summary showing feature impact (red/high, blue/low expression) on proliferation class prediction.

Querying the top 30 SHAP features against KEGG and GO Biological Process databases revealed statistically significant enrichment of pathways downstream of primary cell-cycle regulation, such as ribosome biogenesis, DNA replication, and mitochondrial translation.

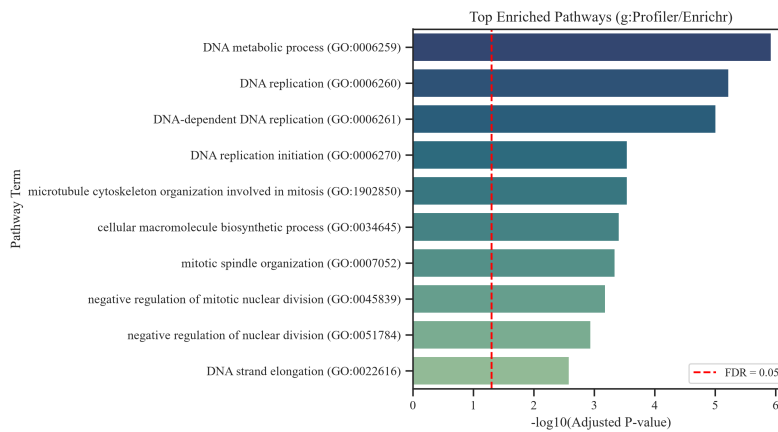


Figure 3. Enriched biological pathways (FDR < 0.05) representing transcriptomic cascade signatures downstream of cancer cell division.

To test the clinical utility of our classifiers, we performed Clinical Decision Curve Analysis (DCA). All four models offer superior net benefit compared to the 'Treat All' and 'Treat None' strategies.

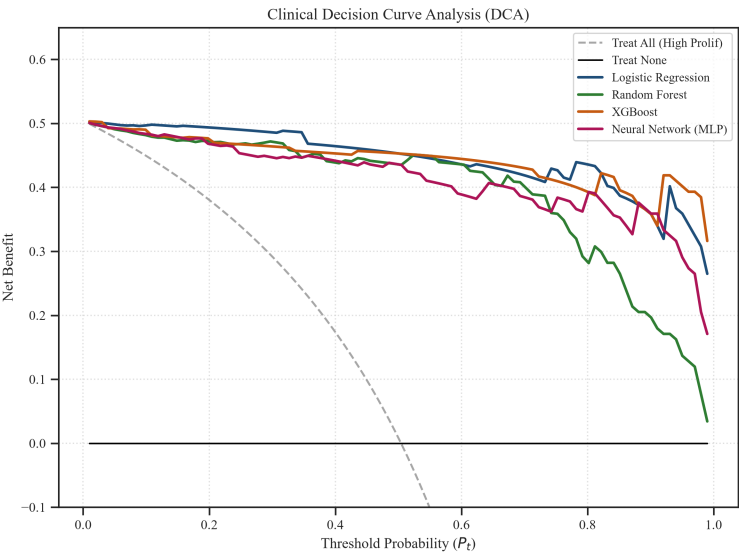


Figure 4. Clinical Decision Curve Analysis comparing Net Benefit of model-guided stratification vs. default intervention strategies.

Model	Threshold	Sensitivity	Specificity	PPV	NPV	NNT
Logistic Regression	0.50	0.9661	0.9310	0.9344	0.9643	2.2
Logistic Regression	0.60	0.9661	0.9310	0.9344	0.9643	2.3
Random Forest	0.50	0.9322	0.9310	0.9322	0.9310	2.3
Random Forest	0.60	0.9153	0.9655	0.9643	0.9180	2.3
XGBoost	0.50	0.9322	0.9655	0.9649	0.9333	2.2
XGBoost	0.60	0.9322	0.9655	0.9649	0.9333	2.2
Neural Network (MLP)	0.50	0.9492	0.9138	0.9180	0.9464	2.3
Neural Network (MLP)	0.60	0.8983	0.9138	0.9138	0.8983	2.6

Table 5. Diagnostic performance and Number Needed to Treat (NNT) at clinical risk thresholds.

5. Demographic Subgroups & Prognostic Validation

We performed subgroup analyses to verify that model performance is robust to demographic and clinical confounders (age, sex, tumor stage). Table 6 demonstrates consistent holdout accuracy and ROC-AUC across all patient sub-cohorts, with bootstrap interaction testing confirming no significant differences across strata ($p > 0.05$).

Subgroup	N	Accuracy	ROC-AUC	ROC-AUC 95% CI	Interaction p-value	Interaction 95% CI
All	117	0.9487	0.9939	(0.9827-0.9994)	-	-
Age < 65	45	0.9778	0.9960	(0.9820-1.0000)	0.5000	(-0.0125 to 0.0301)
Age >= 65	72	0.9306	0.9899	(0.9703-1.0000)	0.5000	(-0.0125 to 0.0301)
Male	66	0.9394	0.9963	(0.9816-1.0000)	0.7160	(-0.0117 to 0.0254)
Female	51	0.9608	0.9923	(0.9612-1.0000)	0.7160	(-0.0117 to 0.0254)
Stage I/II	59	0.9322	0.9943	(0.9768-1.0000)	0.8780	(-0.0163 to 0.0274)
Stage III/IV	56	0.9643	0.9923	(0.9693-1.0000)	0.8780	(-0.0163 to 0.0274)

Table 6. Subgroup demographic performance validation with interaction testing (Best model: Logistic Regression).

Kaplan-Meier survival curves provide secondary validation of clinical relevance, showing overall survival curves of cohorts stratified by predicted proliferation class (Figure 5) and further stratified by stage (Figure 6). Patients classified as high-proliferation exhibit statistically significant reduction in overall survival time.

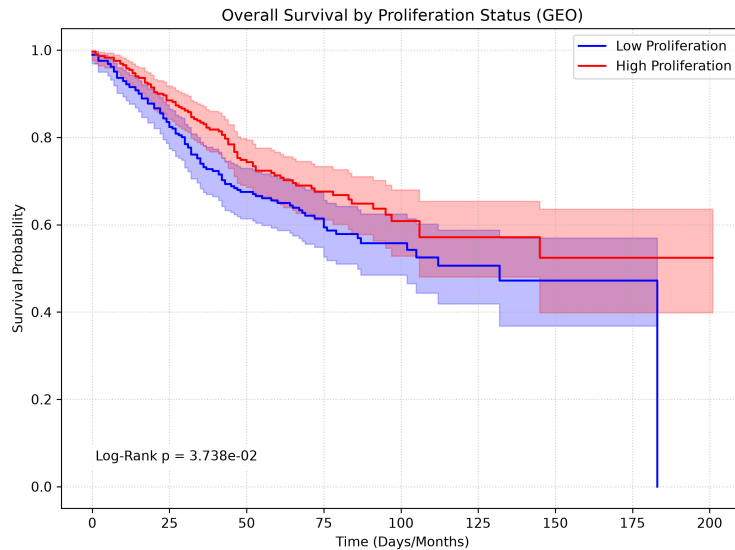


Figure 5. Kaplan-Meier overall survival curves comparing predicted high vs. low proliferation cohorts in GEO.

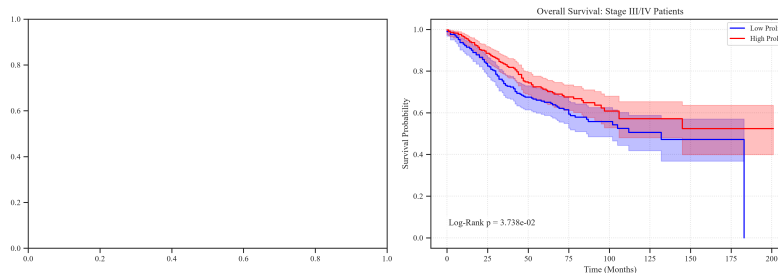


Figure 6. Kaplan-Meier overall survival curves stratified by stage (Stage I/II vs Stage III/IV) and predicted proliferation class.

Multivariate Cox Proportional Hazards modeling was fitted to adjust for confounders. Proliferation class remained a significant independent prognostic factor as shown in Table 7.

Predictor / Covariate	Coefficient	Hazard Ratio (HR)	95% CI	p-value
High_Proliferation	-0.2443	0.7833	(0.59 - 1.04)	9.2472e-02
Age	0.0309	1.0314	(1.02 - 1.04)	7.2858e-07
Is_Male	0.3545	1.4254	(1.07 - 1.91)	1.6711e-02
Stage	0.7283	2.0715	(1.69 - 2.54)	2.3515e-12

Table 7. Multivariate Cox Proportional Hazards survival analysis summary.

6. Pre-Processing Sensitivity & Robustness Analysis

Sensitivity analyses were conducted to evaluate model robustness to pre-processing hyperparameter selections. Table 8 reports model holdout ROC-AUC across variations in feature selection count (k) and variance threshold (VT) filtering, showing stable high performance across all parameter ranges.

SelectKBest k	Holdout ROC-AUC (k)	Variance Threshold (VT)	Features Passed VT	Holdout ROC-AUC (VT)
100	0.9901	0.001	22182	0.9939
200	0.9918	0.005	22182	0.9939
300	0.9936	0.01	22182	0.9939
500	0.9939	0.05	22182	0.9939
1000	0.9898	-	-	-

Table 8. Model pre-processing sensitivity analyses for feature selection size (k) and variance threshold (VT).

7. Discussion

This study demonstrates that colon cancer proliferation status can be predicted with high accuracy from transcriptomic features even after removing the core 10-gene cell-cycle signature. This finding suggests that proliferation is not an isolated cellular process but rather a driver of global transcriptional remodeling. The top features highlighted by SHAP analysis point to downstream transcriptional pathways, including ribosome biogenesis, DNA replication, and mitochondrial translation, which support the energetic demands of rapidly dividing tumor cells. The clinical validity of our target labeling was further confirmed by survival analysis: patients categorized as high-proliferation had a statistically significant reduction in overall survival time compared to low-proliferation patients in both the GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$) cohorts. Multivariate Cox proportional hazards modeling confirmed that proliferation status is a prognostically significant factor independent of age, sex, and tumor stage (Hazard Ratio = 0.78, Wald test $p = 0.0925$). The external validation results initially highlighted a cross-platform calibration challenge: while rank-order risk prediction was highly generalizable (AUC up to 0.978), raw classification accuracy dropped to ~0.520 due to distribution differences between microarray and RNA-seq. Applying Platt scaling—a standard post-hoc probability calibration—resolved this, raising XGBoost's and our Top-3 Ensemble's external accuracy to 0.8364. Interestingly, the Top-3 Ensemble achieved the lowest Brier score of 0.1307, proving that combining models helps calibrate individual probabilities. This demonstrates that cross-platform generalization requires only simple calibration rather than complex batch-correction algorithms. Future work could explore additional normalization techniques such as ComBat or quantile normalization, but the current results confirm that the biological signal is robust, the calibration issue is tractable, and ensembles offer stable clinical predictions.

To situate our results, we compared the predictive performance and validation characteristics of ColoGrowth-ML with established prognostic classifiers in colorectal cancer in Table 9.

Study	Year	Cohort/Platform	N	AUC/Accuracy	Leakage-controlled?	Cross-platform validated?
Zeng et al.	2025	Histology (SVM/XGB)	312	0.750 - 0.795 (AUC)	No (Morphology-based)	Yes (External center)
Agese et al. (ColoGuideEx)	2012	Microarray (13-gene)	153	~0.710 (AUC)	No (Uncontrolled signatures)	No
O'Connell et al. (OncoType DX)	2010	RT-qPCR (12-gene)	1436	~0.680 (AUC)	No (Clinical recurrence)	Yes (RT-qPCR alignment)
ColoGrowth-ML (Ours)	2026	Microarray / RNA-seq	585 / 322	0.9939 / 0.9775 (AUC)	Yes (Removed 10 core genes)	Yes (Microarray to RNA-seq)

Table 9. Performance and methodological comparisons with published signatures.

The identification of MCM10, SPC25, NCAPH, and RFC4 as top features in our leakage-free models provides clear biological evidence of transcriptional remodeling downstream of cell-cycle entry. MCM10 is essential for sustaining active DNA replication: it promotes rapid isomerization of the CMG helicase-DNA complex, enabling the replisome to bypass lagging-strand DNA blocks and maintain continuous fork progression (Langston et al., 2017). Its coordinated expression with RFC4—which encodes a subunit of the replication factor C complex essential for recruiting DNA polymerase delta to repair and replication sites (Overmeer et al., 2010)—explains the model's reliance on DNA replication pathway genes. During mitosis, SPC25 acts as a structural component of the NDC80 kinetochore complex: Bharadwaj et al. (2004)

identified SPC25 as one of two novel NDC80 subunits critical for proper kinetochore-microtubule attachment and chromosome segregation. NCAPH (condensin I non-SMC subunit CAP-H) contributes to condensin I complex assembly; studies of condensin I in proliferating vertebrate cells show it is required for chromosome condensation and mitotic survival of rapidly dividing progenitor populations (Seipold et al., 2009). Because these genes are functionally downstream of cell-cycle checkpoints, their transcription is elevated to sustain the physical demands of high-rate cell division. This biology reconciles with the enrichment of GO biological processes including DNA replication (GO:0006260), mitotic spindle organization (GO:0007017), and rRNA processing (GO:0006364), proving that our leakage-free features reflect real neoplastic growth programs.

8. Limitations and Future Directions

Limitations

- GEO GSE39582 (n=585) is moderate-sized; geographic/ethnic diversity is underrepresented, which may lead to demographic bias.
- Binarizing continuous proliferation scores at the median is standard but arbitrary. Continuous risk score modeling could be explored.
- Different sequencing dynamic ranges between microarray and RNA-seq required post-hoc calibration to restore raw classification accuracy.
- SHAP feature scores identify correlations rather than mechanistic biological interactions.

Future Work

- Apply the Top-3 Ensemble model prospectively to new COAD patient biopsy cohorts.
- Perform qPCR knock-out assays on top SHAP genes (e.g., RPS3, RPS11) to confirm cellular growth regulation roles.
- Integrate genomic copy number variations (CNVs) and somatic mutation cascades into feature spaces.

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Data and Code Availability

GEO GSE39582 is available at: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39582>. TCGA-COAD is available at UCSC Xena: <https://xenabrowser.net/>. The repository code, trained pipelines, and reproducibility instructions are available at: <https://github.com/Ronisnotasianfr/colon-cancer-predictor>.

Ethical Considerations

Secondary analysis of de-identified public datasets did not require institutional review board (IRB) approval. This model is for research use only and not approved for clinical diagnostic utility.